

Corrected Document Output

Generated on: 2025-06-13 21:21:38

Chapter 15 Imperialist Competitive Algorithm Abstract In this chapter, we present a new optimization algorithm called imperialist competitive algorithm (ICA), which is inspired by the human socio-political evolution process. We first describe the general knowledge of the imperialism in the United States. Sect. 15.1. Then, the fundamentals and performance of ICA are introduced in Sect. 1.15.2. Finally, 1st section, 15.3, summarises this chapter. 15.1 Introduction to the 'empire' In politics and history, in order to explain the extending control over weaker people or areas, the term "empire" is appeared. One of the famous examples is the reign of the Greco-Roman Empire from 27 BC to 476. The most powerful person is usually called an imperialist and different imperialists will extend their authority and control of one state or people over another through the form of competition. Inspired by this human socio-political evolution process, a newly developed evolutionary algorithm called Imperialist competitive algorithm (ICA) is proposed by Atashpaz-e-Gargari and Lucas (2007). 15.1.1 Imperialism Imperialism was greatly influenced by an economic theory known as mercantilism, which inspired the government to extend their power and the rule beyond its own boundaries (Atashpaz-Gargari and Lucas 2007). In its initial forms, due to the limited supply of wealth, countries focused on building their empire with natural resources, such as gold and silver. However, nowadays, the new imperialism focused on the new technological advances and developments. The ultimate goal is to increase the number of their colonies and spread their empires over the world. B. Xing and W.-J. Gao, Innovative Computational Intelligence: , Â d , , . 203 pp. A Rough Guide to 134 Clever Algorithms, Intelligent Systems Reference Library 62, No. 1 (2014). DOI: 10.1007/978-3-319-03404-1_15, _____ Copyright © 2012-2017 Springer International Publishing Switzerland 2014

15.2.1 Fundamentals of Imperialist Competitive Algorithm 15.1.1 Â d ch , m . . 15
Imperialist competitive Algorithm (ICA) 15.3.1 Fundamentals of 15.4.1
IMFamentals 15.5.2 Imperialist Competition Algorithm 15.5-15.6 IMF 15.7
IMF 15-15-2.2 IMF Imperialist competitive algorithm 15-5.1 Imperialist Competitive algorithm (ICAI) was originally proposed by At In ICA, the countries can be viewed as population individuals and basically divided into two groups based on their power, i.e., colonies and imperialists. Also, one empire is formed by one imperialist with its colonies. Furthermore, two operators called assimilation and revolution and one strategy called imperialistic competition are the main building blocks that employed in ICA. The implementation procedures are described below. • Initializing phase: 1.

Preparation of initial populations. Each solution (i.e., country) that in form of an array can be defined as follows: country $\frac{1}{4} \frac{1}{4} p_1; \dots$ an array. The country of a country is the same as a country. The country of the array is defined via Eq. 15.1 (Atashpaz-Gargari and Lucas 2007): country $\frac{1}{4} \frac{1}{2} p_1; p_2; \dots$. Itude false, pnvat, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 15, 16, 17, 18, 19, 20, 21, 22, 29, 25, 30, 32, 39, 39. 20, 32. 32.0% 1.0(2.0). This represent different variables which, based on various socio-political characteristics (such as culture, material, and economical policy) and Nvar, denotes the total number of the characteristics (i.e., n-dimensional of the 2. Creating the cost function. In order to evaluate the cost of countries, the cost-of-function can be defined via Eq. 15.2 (Atashpaz-Gargari and Lucas 2007):... The cost of a country is: $\cos t \frac{1}{4} f \delta$ country $P \frac{1}{4} F \delta p_1; p_2; \dots$. Itude 1.0.5.5, pnvat 1.2, 1.1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 15, 16, 20, 21, 22, 29, 30, 32, 26, 32.0%2.2.1. Initializing the empires: In general, the initial size of populations (Npop) and the size of empires (Np) involves two types of countries [i.e., colony (Ncol) and imperialist (Nimp)] which together form the empires. To form the initial empires proportionally, the normalized cost of an imperialist is defined via Eq. 15.3.3 (Atashpaz-Gargari and shroud shrouds, 2007), $\text{purseguarding, } cn \text{ maxi } f \text{ ci gn, } 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 21, 22, 20 \text{ and } 21, 20.6$. The difference is that in New Zealand researchers took the necessary steps and began treatment. citation The treatment seemed to be effective on the pups who have taken it. Normally, two methods can be used to divide colonies among imperialists: (1) from the imperialists' point of view which based on the power of each imperialist; (2) from the colonies' point of view which based on the relationship with the imperialist (i.e., the colonies should be possessed by the imperialist according to the power). Both methods are given via Eqs. 15.4 and 15.5, respectively (Atash-paz-Gargari and Lucas 2007).

Itude 15.2.2 Imperialist Competitive Algorithm 8.1, 8.2, 9.5, 8, 9, 11, 12, 13, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 29, 28, 34, 35, 39, 36, 38, 39.6, 38.6% of the stored produce has already rotted. I think that the Union' s natural role would be to dedicate itself to this, and I hope we can build on this report to make Parliament and the Commission aware of the need for intervention. • Moving phase: 1.1. Itude Assimilation strategy. According to this strategy, all colonies will move toward maximizing their relevant imperialist with x units via Eq. 15.6.6 (Atashpaz-Gargari and Lucas haircut haircut,2007), $\text{lacklusterx, } 1.5 U \delta, 0.6, 2.5, 3.0, 4.4, 5.5 6.0 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 21 \text{ and } 21$. The difference is that in New Zealand researchers 2. Revolution strategy: According to this strategy, a random amount of deviation from the direction of movement is incorporated via Eq to the direction of movement. Itude 15.7.7 (Atashpaz-Gargari and Lucas, 2007) Courthouse, Chogh U. Û ð C.L.C. 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 12, 13, 14, 15, 16, 16 and 16, 12.0 ð15.5.0.2.2, where h is a random variable with uniform distribution, and c is a

parameter that adjusts the deviation from the original direction. • • • • Based on the cost function, when the new position of a colony is better than that of the corresponding imperialist, the imperialist and the colony change their positions and the new location with a lower cost becomes the imperialist. • Imperialistic competition phase: 1-1. Calculating the total power of an empire: It is influenced by the power of the states of an anti-imperialist country and the colonies of an imperial empire via Eq. 15.8 (Atashpaz-e-Tashkent) of the $ITC_n = \frac{1}{4} \text{Cost} \delta \text{colonies of empire } n + p_n \text{ mean } f_{\text{cost} \delta \text{empires of empire } n} + g$; ■ $\delta 15:8p$ where TC_n is the total cost of the n th empire, and n is a positive number which is considered to be less than 1.1.2. Imperialistic competition strategy. According to this strategy, all empires try to take control of the colonies of other empires and control them. To.

206, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 22, 24, 25, 29, 28, 34, 33, 35, 39, 38, 37, 38. 20, 39. 23, 37. Imperialist Competitive Algorithm modelled this strategy. The weakest colonies of the weakest empires will be forced to compete among all other empires in order to possess this colony. Ance Based on TCN , the normalized total cost is simply obtained via Eq. 15.9%-9.5%-6.5% $NTCN$, 1.2% $TTCN$ 1.5, 2.6, 3.6% 2.7% 3.4% 4.5%, 5.6%, 6.6%.6.0% 5.5%.5.1% 6.5($TPCN$ represents the total normalized cost of the empire. Having $NTCN$, the possession probability of each empire is evaluated via Eq. 15.10.10

(Atashpaz-managed $NTCNnPINPINpPnPINPpnPQ$, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 22, 24, 25, 29, 30, 34, 35, 39, 38, 41, 38. 20, 39. 10, 41. 15, 37, 39 6, 38 8, 39 3, 39 8, 40, 45, 38 or 39. 20% of Eliminating phase: When an empire loses all its colonies (i.e., their colonies will be divided among other empires), it is assumed to be collapsed and will be eliminated. • Convergence phase: At the end, all the colonies will be under the control of the most powerful empire, which means all the colonies have the same positions and costs and will be controlled by an imperialist with the same position and cost as themselves. In other words, there are no differences not only between colonies, but also between colonies and imperialists. Taking into account the key phases described above, the steps of implementing ICA can be summarized as follows (Atashpaz-Gargari and Lucas 2007): • Step 1: Defining the optimization problem. Left Step 2.1.2. Generating initial empires by pick some random points on the function. • Step 3: Move the colonies towards imperialist states in different directions (i.e., through the process of re-assimilation). • Step 4: Random changes occur in the characteristics of some countries (i.e., the result of a counter-revolution). • Step 5: Position exchange between a colony and an imperialist. Step 6: Calculate the total cost of all empires. • Step 7: Use imperialistic competition and pick the weakest colony from the next-weakest empire. • Step 8: Eliminate the powerless empires. 1st Step 9.0.1. Check if maximum iteration is reached, go to Step 3 for new beginning. If a specified termination criteria is satisfied, stop and return the

best solution.

15.2.2 Imperialist Competitive Algorithm 1, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 16, 17, 18, 19, 20, 21, 22, 24, 25, 28, 29, 34, 35, 38, 39, 38.2, 39.5, 45.6% of 32.5% of 16.2% performance of ICA. In order to show how the ICA performs, four minimization problems are tested in this study: Atashpaz-Gargari and Lucas (2007). Compared with genetic algorithm (GA) and particle swarm optimization (PSO), the results showed that ICA is capable of reaching the global minimum. 15.3 Conclusions and Recommendations In this chapter, an optimization algorithm motivated by one of the socio-political models (i.e., imperialistic competition) is introduced. The term "countries" are designed as the initial populations and categorized into colony and imperialist states, respectively. Both colonies and imperialists together form the empires. The basic idea behind ICA is to lead the search process toward the powerful imperialist or the optimum points based on their 'power', i.e., the weakest empires will lose their colonies until there will be no colony in that. The objective of ICA is to find an ideal world in which there is no difference not only between colonies but also between colonies and imperialists. Although it is a newly introduced computational intelligence method, we have witnessed the following rapid spreading of ICA: First, several enhanced versions of ICA can be found in the literature as out-of-print code below: • Bacterial foraging optimization based on ICA (Acharya et al. 2010). • Left Chaotic improved ICA (Talatahari et al. 2012a, 2012a and 2012a). • Constrained genetic algorithm based ICA (Acharya et al. 2010). • Report Fast ICA (Acharya et al. 2006). 2010). • Diagnostic Hybrid artificial neural network and ICA (Taghavifar et al. 2013). • Hybrid evolutionary algorithm and ICA (Ramezani et al. 2012). • Modified ICA (Niknam et al. 2011). • Project Multiobjective ICA. (Bijami et al. 2011; Mohammadi et al. 2011). • • • Quad countries algorithm (Soltani-Sarvestani et al. 2012). Second, the ICA has also been successfully applied to a variety of optimization problems, as listed below, minimizing artificial neural network training (Moadi et al. 2006). 2011) Ance. • Assembly line balancing (Bagher et al. 2011). • Data mining (Ghanavati et al. 2011; Niknam et al. 2011). • Facility location optimization (Mohammadi et al. 2011; Moadi et al. 2011). • Power system optimization (Nejad and Jahani 2011; Bijami et al. 2011). • Product mix-outsourcing problem (Nazari-Shirkouhi et al. 2010). • Scheduling optimization (Forouharfard and Zandieh 2010; Ayough et al. 2012; Lian et al. 2012; Kayvanfar and Zandieh 2012).

208, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 22, 24, 25, 27, 28, 29, 34, 35, 38, 39, 38. 20, 39. Imperialist Competitive Algorithm, JPEG, Soil computation prediction (Taghavifar et al., 2013). • Structure design optimization (Talatahari et al. 2012b). Interested readers please refer to them as a starting point for further exploration and exploitation of ICA. References Acharya, D. P., Panda, G., &

Lakshmi, Y. V. S. (2010). Effects of finite register length on fast ICA, bacterial foraging optimization based ICA and constrained genetic algorithm based ICA algorithm. *Digital Signal Processing*, 20, 964-975.

Atashpaz-Gargari, E., & Lucas, C. (2007). Imperialist competitive algorithm: An algorithm for optimization inspired by imperialistic competition. In *IEEE Congress on Evolutionary Computation and Inter-Computation (CEC 2007)* (pp. 4661-4667). IEEE.

Ayough, A., Zandieh, M., & Farsijani, H. (2012). GA and ICA approaches to job rotation scheduling problem, considering employee's boredom. *International Journal of Advanced Manufacturing Technology*, 60, 651-666.

Bagher, M., Zandieh, M., & Farsijani, H. (2011). Balancing of stochastic U-type assembly lines: an algorithm. *International Journal of Advanced Manufacturing Technology*, 54, 271-285.

Bijami, E., Abshari, R., Askari, J., Hosseinnia, S., & Farsangi, M. M. (2011). Optimal design of damping controllers for multi-machine power systems using metaheuristic techniques. *International Review of Electrical Engineering*, 7, 6, 1883-1894.

Forouharfard, S., & Zandieh, M. (2010). An algorithm to apply a competitive algorithm to schedule of receiving and shipping trucks in cross-docking systems. *International Journal of Advanced Manufacturing Technology*, 51, 1179-1193.

Ghanavati, M., Gholamian, M. R., Minaei, B., & Davoudi, M. (2011). An efficient cost function for competitive algorithm to find best clusters. *Journal of Theoretical and Applied Information Technology*, 29, 22-31.

Kayvanfar, V., & Zandieh, M. (2012). The economic lot scheduling problem with deteriorating items and shortage of an imperialist competitive algorithm. *International Journal of Advanced Micro-Manufacturing Technology*. Doi:10.1007/s00170-011-3820-6.

Lian, K., Zhang, C., Shao, X., and Gao, L. (2012). Optimization of process planning with various flexibilities using an imperialist competitive algorithm. *International Journal of Advanced Manufacturing Technology*, 59, 815-828.

Moadi, S., Mohaymany, A. S., & Babaei, M. (2011). Application of a competitive algorithm to the emergency medical services location problem. *International Journal of Artificial Intelligence and Applications (IJAIA)*, 2, 137-147.

Mohammadi, M., Tavakkoli-Moghaddam, R., and Rostami, H. (2011). A multi-objective competitive algorithm for a capacitated hub covering location problem. *Nazari-Shirkouhi, S., Eivazy, H., Ghodsi, R., Rezaie, K., & Atashpaz-Gargari, E. (2010). Solving the integrated product mix-outsourcing problem using the imperialist competitive algorithm. Expert Systems with Applications, 37, 39, 38, 49, 76, 77, 76-7626.*

Nejad, H. C., & Jahani, R. (2011). A new approach to economic load dispatch of power system using imperialist competitive algorithm. *Australian Journal of Basic and Applied Sciences*, 5, 835-843.

Niknam, T., Fard, E. T., Ehrampoosh, S., & Rousta, A. (2011). A new hybrid competitive algorithm on data clustering. *Sa dhana*, 36, 293-315.

References 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 22, 24, 27, 28, 29, 35, 38, 39, 38. 39, 36, 37, 41, 45, 46, 49, 39. 45, 45. 46, 46. 47, 49. 48. 49, 49(47). A hybrid evolutionary imperialist competitive algorithm (HEICA). In J.-S. Pan, S.-M. Chen, N. T. Nguyen (Eds.) ACIIDS, 2012, Part I, LNAI 7196 (pp. 359-368). Soltani-Sarvestani, M. A., Lotfi, S., & Ramezani, F. (2012). Quad Cities algorithm (QCA). In: A.J.-S. Pan, S.-M. Chen & N. T. Nguyen (Eds.) ACIIDS 2012, Part III, LNAI 7198 (pp. 119-129). Berlin: Springer. Taghavifar, H., Mardani, A., & Taghavifar, L. (2013). A hybridized artificial neural network and competitive algorithm optimization approach for prediction of soil compaction in soil bin facility. *Measurement*, 46, 2288-2299. Talatahari, S., Azar, B. F., Sheikholeslami, R., & Gandomi, A. H. (2012a). Imperialist competitive algorithm combined with chaos for global optimization. *Communications, Insurrection, Nonlinear Science, and Numerical Simulation*, 1, 17, 19, 20, 21, 24, 24-25, 21-25. Talatahari, S., Kaveh, A., & Sheikholeslami, R. (2012b). Chaotic iphona is a competitive algorithm for optimum design of truss structures. *Structural and multidisciplinary optimization*. Doi:10.1007/s00158-011-0754-4.